MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

Artificial Intelligence and Society

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Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 4.1: Text as Data

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Das Interdisziplinäre Digitale
Labor der Universität Graz

Converting Text to Numbers

- There are around 600,000 words in the Oxford English Dictionary.
- Representing text as one-hot encoded vectors is extremely inefficient.
- How can we make the encoding more efficient?
- **Solution**: learned "embeddings".

	1	2	3	4	5	6	7	8
cat	1	0	0	0	0	0	0	0
kitten	0	1	0	0	0	0	0	0
dog	0	0	1	0	0	0	0	0
houses	0	0	0	1	0	0	0	0
man	0	0	0	0	1	0	0	0
women	0	0	0	0	0	1	0	0
king	0	0	0	0	0	0	1	0
queen	0	0	0	0	0	0	0	1

word — *one-hot ecoding*

each word is represented by a vector with 8 entries

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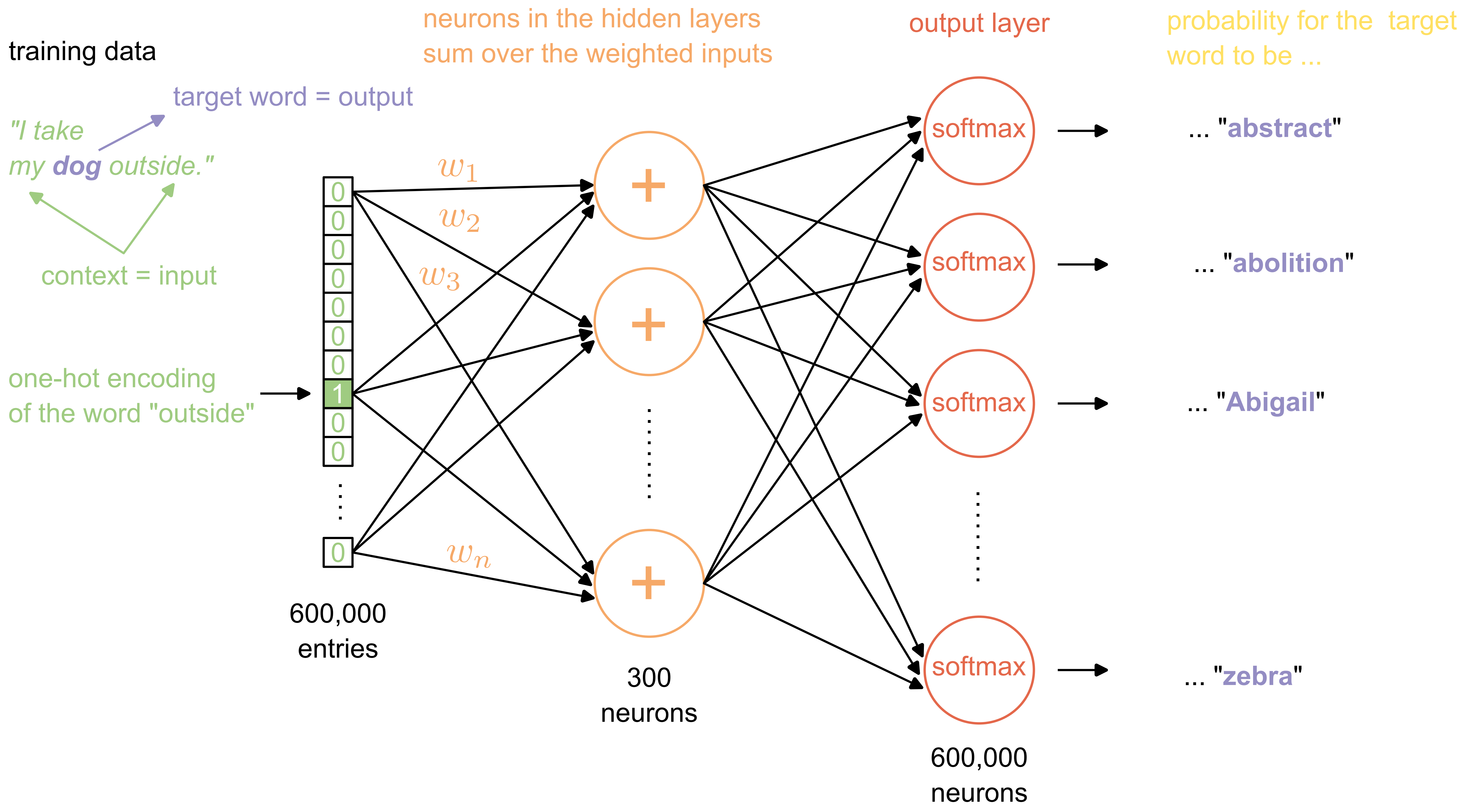
	living being	cat-like	human	gender	king-like	verb	plural
cat	0.6	0.9	0.1	0.4	-0.7	-0.3	-0.2
kitten	0.5	0.8	-0.1	0.2	-0.6	-0.5	-0.1
dog	0.7	-0.1	0.4	0.3	-0.4	-0.1	-0.3
houses	-0.8	-0.4	-0.5	0.1	-0.9	0.3	0.8
man	0.6	-0.2	0.8	0.9	-0.1	-0.9	-0.7
women	0.7	0.3	0.9	-0.7	0.1	-0.5	-0.4
king	0.5	-0.4	0.7	0.8	0.9	-0.7	-0.6
queen	0.8	-0.1	0.8	-0.9	0.8	-0.5	-0.9

word *embedding*

each word is represented by a vector with ***7 entries***

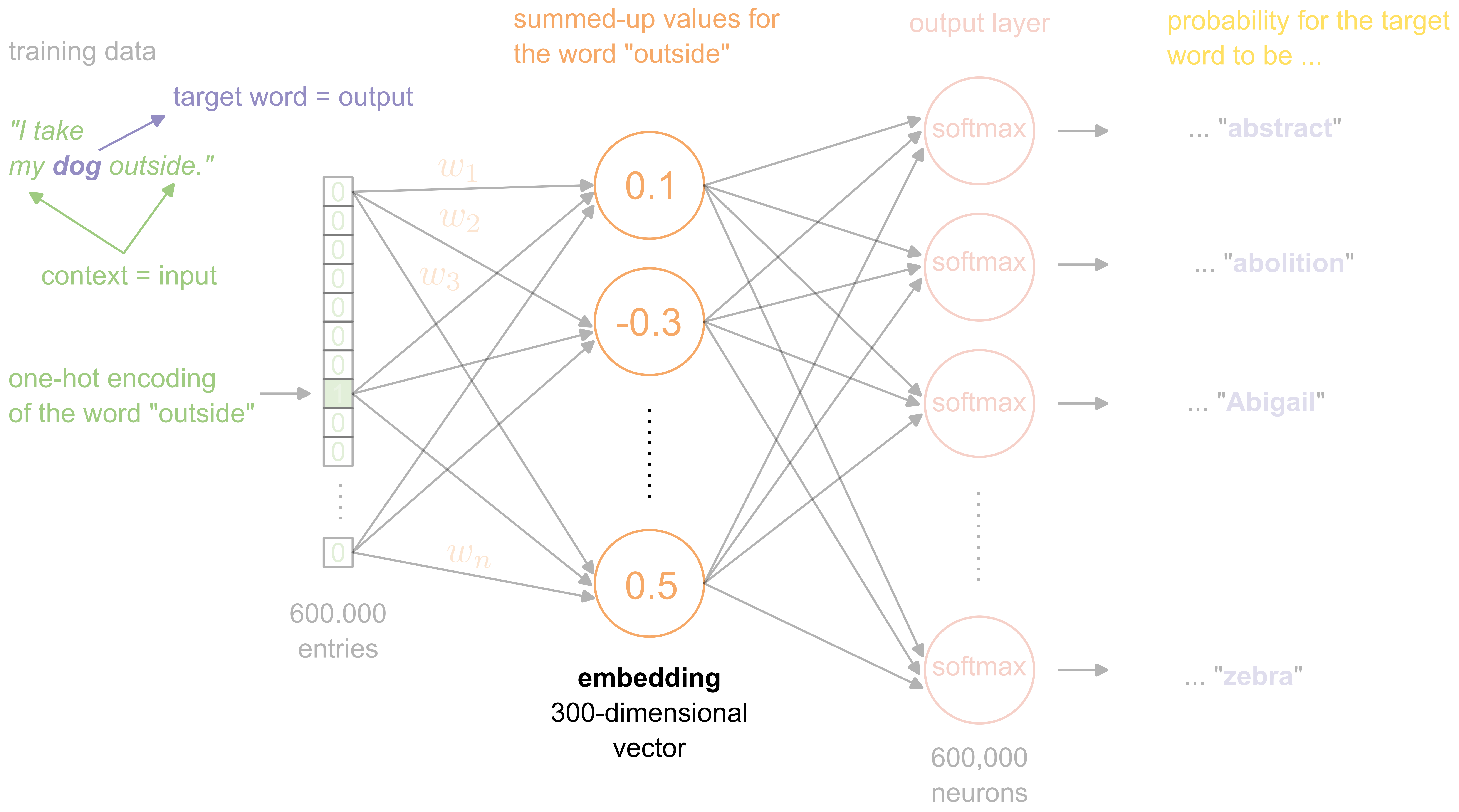
adapted from
Word embedding: basics"
Hariom Gautam @Medium.

Learning Embeddings



Continuous Bag-Of-Words (CBOW) predicts the target-word based on its surrounding words. Inspired by Manan Suri, A Dummy's Guide to Word2Vec, *Medium*.

Learning Embeddings



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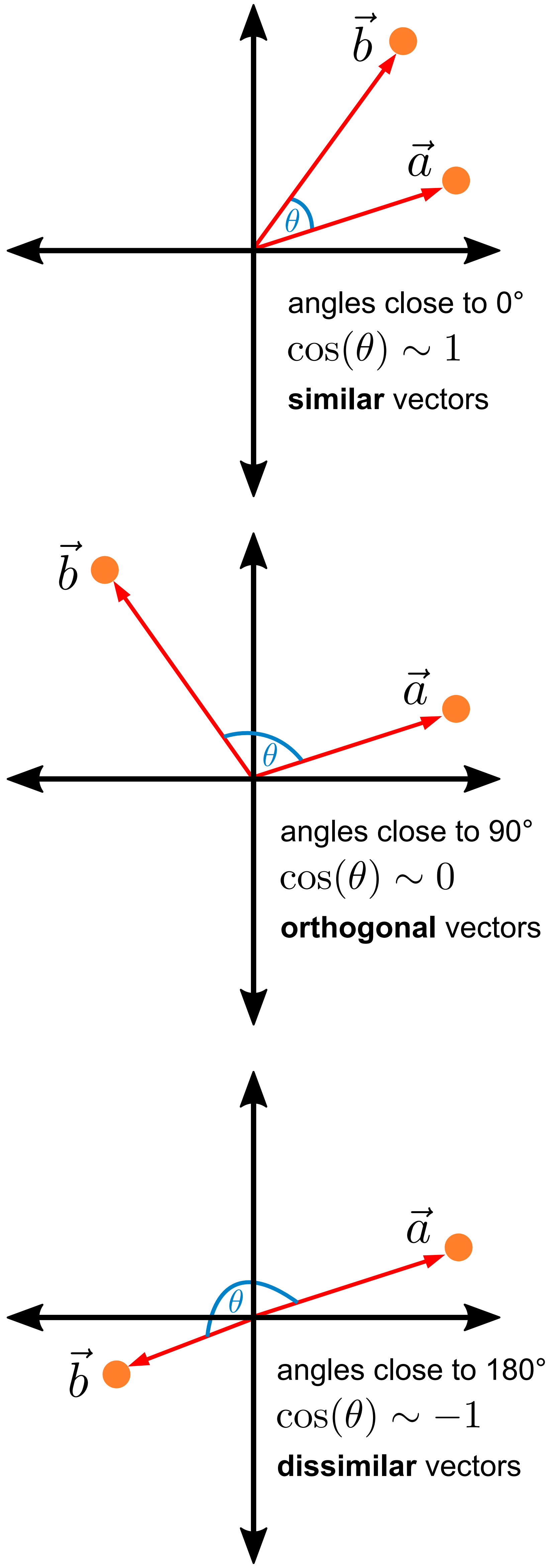
Advantages of Embeddings

- We need far fewer dimensions (typically between 100 and 1000) to represent a word.
- Words that are similar to each other are located at similar positions in the *embedding space*. The **cosine similarity** of their embedding vectors is very high.

$$S_C(\vec{a},\vec{b}) := \cos(\theta) = \frac{\vec{a}\cdot\vec{b}}{||\vec{a}||\ ||\vec{b}||} = \frac{\sum_{i=1}^{n} a_i b_i}{\sqrt{\sum_{i=1}^{n} a_i^2}\cdot\sqrt{\sum_{i=1}^{n} b_i^2}}$$

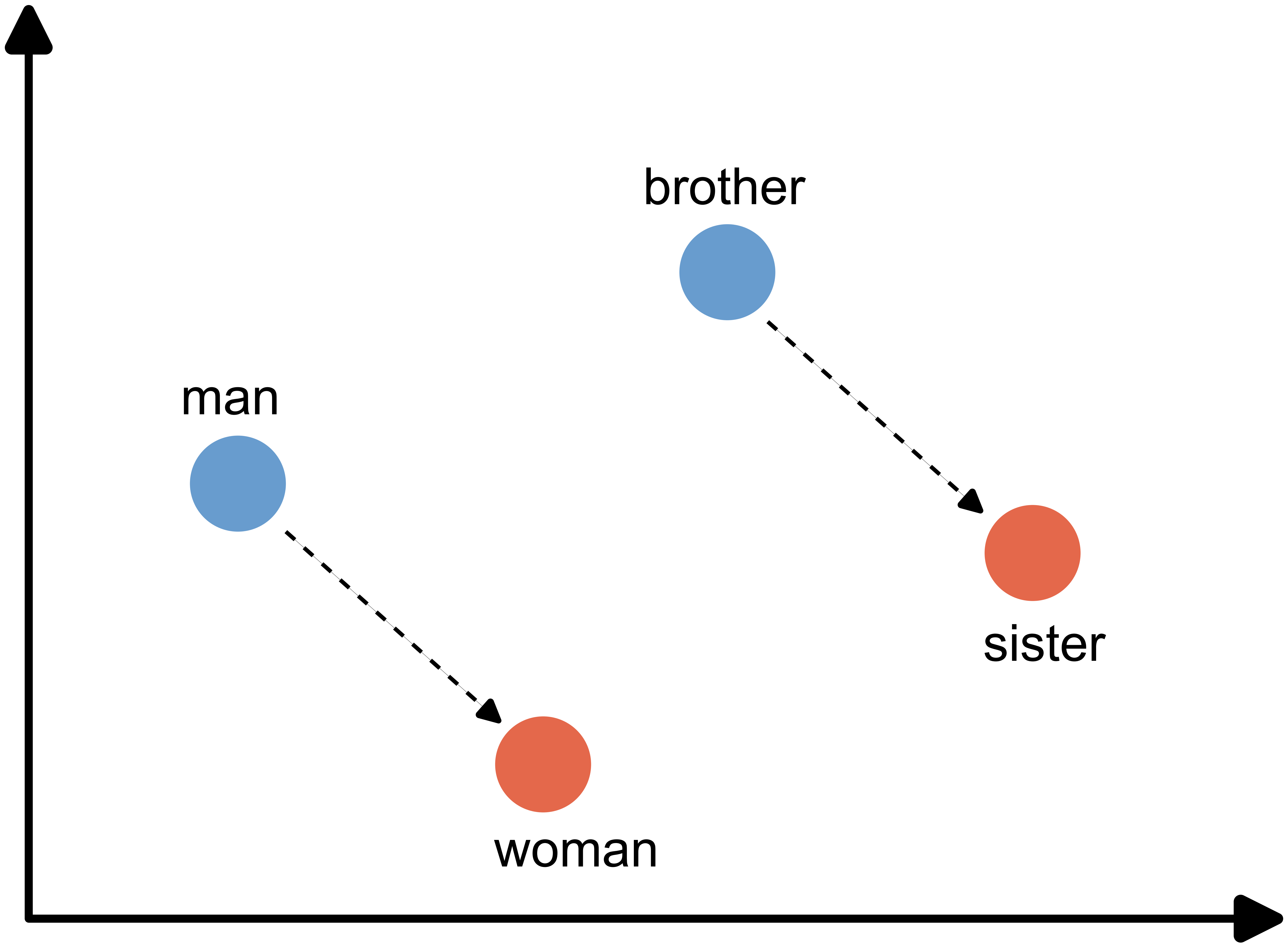
cosine similarity

- We can "calculate" with embedding vectors.



Calculating with Embeddings

"Man" relates to "Woman" in the vector space as "Brother" relates to "Sister".



Summary

- To use text as data for machine learning, we first need to convert it into numbers.
- This is done using **embeddings** - vectors with 100 to 1000 dimensions that represent a word in the context of its language.
- Embeddings are learned using neural networks and large amounts of naturally occurring text.
- The embedding vectors of words that have similar meanings show high cosine similarity - the words are located at similar points in the embedding space.